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Numerical Simulation of several Impact Attenuator Design for a Formula Student Car

Farlian Rizky Sinaga[1], Ubaidillah[1,2,a)], Krishna Eka Kurniawan[1], Muhamad Ivan Fadhil[1], Sukmaji Indro Cahyono[1], and Muhamad Hafiz Idris[3]

[1]*Mechanical Engineering Department, Universitas Sebelas Maret, Surakarta, Jawa Tengah, Indonesia 57126,*
[2] *National Center for Sustainable Transportation Technology (NCSTT), Bandung, Indonesia*
[3]*Malaysia-Japan International Institute of Technology, Universiti Teknologi Malaysia, Jalan Sultan Yahya Petra, Kuala Lumpur*

[a)]Corresponding author: ubaidillah_ft@staff.uns.ac.id

ABSTRACT. In the Formula Society of Automotive Engineer (SAE), safety is a vigorous factor. One of the safety components in the Formula SAE car is the impact attenuator. The purpose of this study is to get the impact attenuator design with the best ability to absorb kinetic energy from several existing designs, through numerical approaches, for estimating conditions against dynamic impacts. Material of impact attenuator use combination of aluminum and Zirconium G350. The simulation was caried out by crashing the impact with the rigid wall, to find the deformation that occurs and the energies are absorbed. The impact attenuator design to be simulated should be optimized to meet some parameters in the SAE Formula. The result of impact attenuator simulation should be able to absorb energy of 7350 joules at move 7 m/s and deformation at bulkhead less than 25.4 mm.

INTRODUCTION

The safety factor is critical for a vehicle, mainly when an accident occurs [1,2]. Driver safety issues on formula vehicles are becoming critical topics [2,3]. Impact attenuators are used on vehicles to reduce the effect of damage to the vehicle during an accident. Good impact attenuator design should be able to absorb the impact of accidents on race cars. [4]. Then the use of impact attenuator is a must-have on the formula car. The higher the capability of a formula car, should also be accompanied by the development of the safety device of the vehicle. Impact attenuator is one of the most critical security devices in the event of an accident so that no damage can harm the driver of the formula car [5].

Vehicle use impact attenuator to reduce the harmful influence to the driver [6]. Various models of the impact attenuators use to minimize the risk of affected when the vehicles that have high velocity [7]. The impact attenuators built on the race cars is also to prevent if the vehicle uncontrollable during an accident [8]. Each model of impact attenuator used must comply with the latest SAE Formula rules, for use on race cars [9]. Of all the methods used for retrieving data in journals, the most widely used way is to perform numerical simulations on various software, both static and dynamic.

This article discusses impact attenuator models that have a good impact capability according to the rules on Formula Student SAE car. After the modeling completed, all model of impact attenuator simulated used dynamics simulation. The numerical simulation uses ANSYS software. In this software, every model that has been in the design will be simulated drove with a certain speed then bump a rigid wall. Each Impact attenuator moves against a wall at a velocity of 7 m/s, which is by the rules on Formula Student SAE. From the simulation will be performed a numerical analysis to find the best model as impact attenuator that will use on FSAE cars.

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COMPUTATIONAL DOMAIN AND BOUNDARY CONDITIONS

This project builds on the analysis of the impact attenuator model that has been studied in advance, to create the characteristics of the results. One of the task to the rules of using the Impact attenuator in the SAE Formula Student is that the most important is construction solution between the attenuator and the expanse barrier: "Attached safely and directly to the Front Bulkhead rather than by being part of the nonstructural bodywork." The Attenuator attachment impact must be built to provide the right path not to be separated by frame, videlicet welded, at least four (4) 8 mm Bolt Grade 8.8 (5/16 inch Grade 5) must attach the Impact attenuator to the Front Bulkhead "(SAE 2017 Formula SAE Rules). In addition to the Impact attenuator shape to be aware of, Impact attenuator mounting modeling also affects the impact of damage to be transferred from the Impact attenuator to the frame. If the transfer of energy received by the frame significantly, it will make the frame accept the impact of the damage, then the Impact attenuator fails.

This simulation purpose to study of impact attenuator properties and characteristics. The three models are defined using the same type of material on each part, that has the same functionality. Material Aluminum 2 mm for Impact attenuator cover, aluminum 4 mm for Anti Intrusion Plate, and STKM 11A for the frame material. Frame represented by frame pieces on Bulkhead with 1.6 mm thick pipe which has a radius of 1 inch (25.4 mm). All models of the impact attenuator design shown in Figures 1.

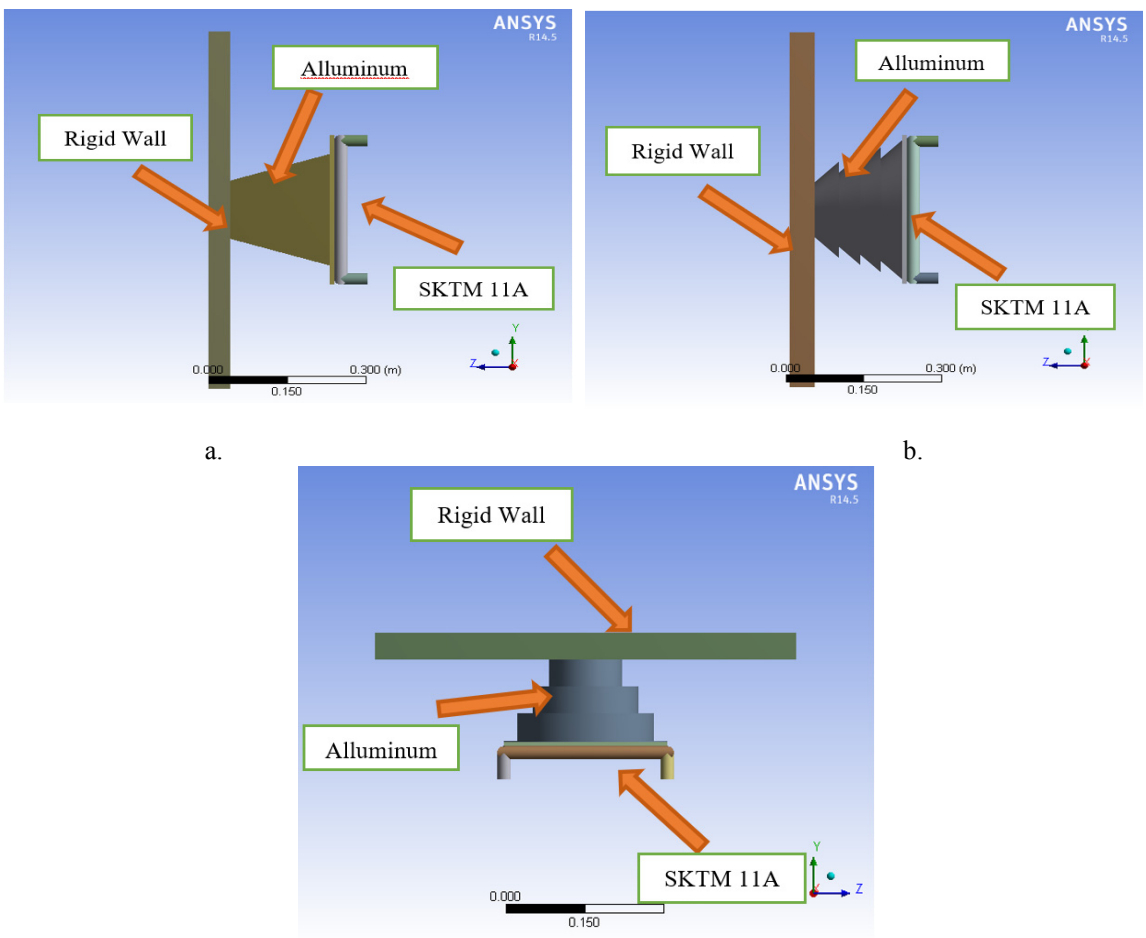


FIGURE 1. Deformation of Impact Attenuator: (a) Model 1, (b) Model 2, and (c) Model 3

All models of impact attenuator have the same mounting method to the frame, and the whole model dimension is the same. All models have the same basic shape, videlicet the size with the most significant tray area lies on the base (the closest part of the frame), while the tray away from the fraction of the cross-sectional area decreases. Differences in the cross-sectional area changes in the distance of the frames on each model are expected to give different effects. The difference between the three models is the shape of the aluminum cover used. The first model resembles a pyramid with a square-shaped top, a second model resembling a pattern on a pine tree, and on a third model of a different diameter cylindrical arrangement, with the most significant diameter as a base connected to the Anti Intrusion Plate. Each model is expected to produce different damage patterns on the Impact attenuator.

For all design of Anti Intrusion Plate and frame, pieces have the same shape and size on each model. The use of frame pieces on each model to see the impact of damage received by impact attenuator will damage the frame or not. Because in the Formula Student SAE rules, bulkhead frame frames should only be maximum 1 inch (25.4 mm) incompressible.

FINITE ELEMENT ANALYSIS PROCEDURE

The calculations in this simulation use several stages to get the proper data to reference. The entire stage of the simulation flowchart of the process shown in Figure 2.

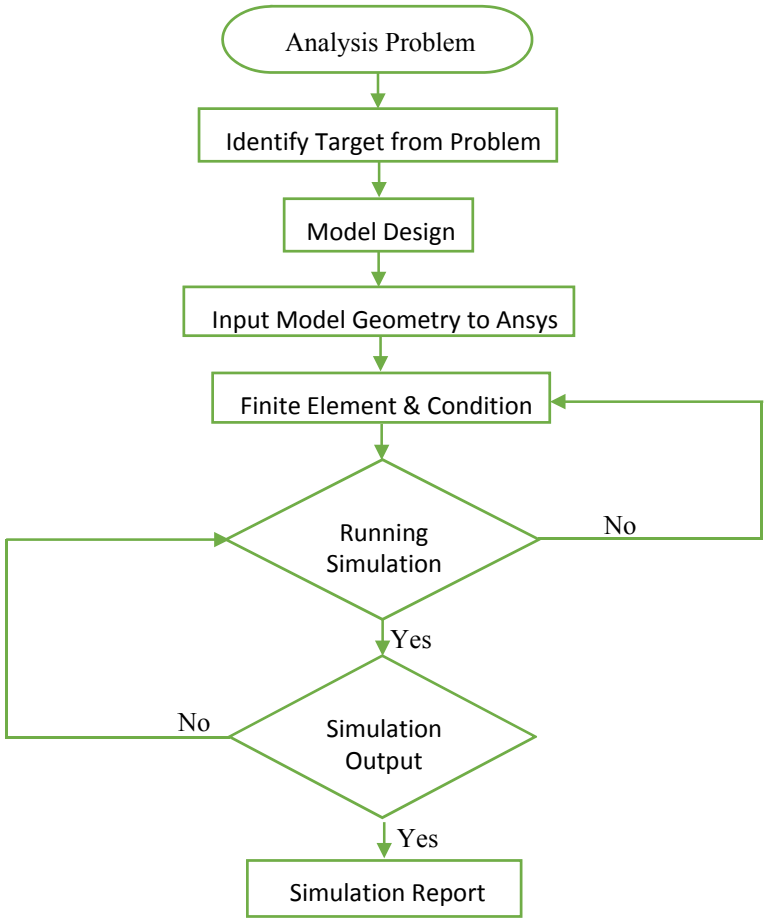
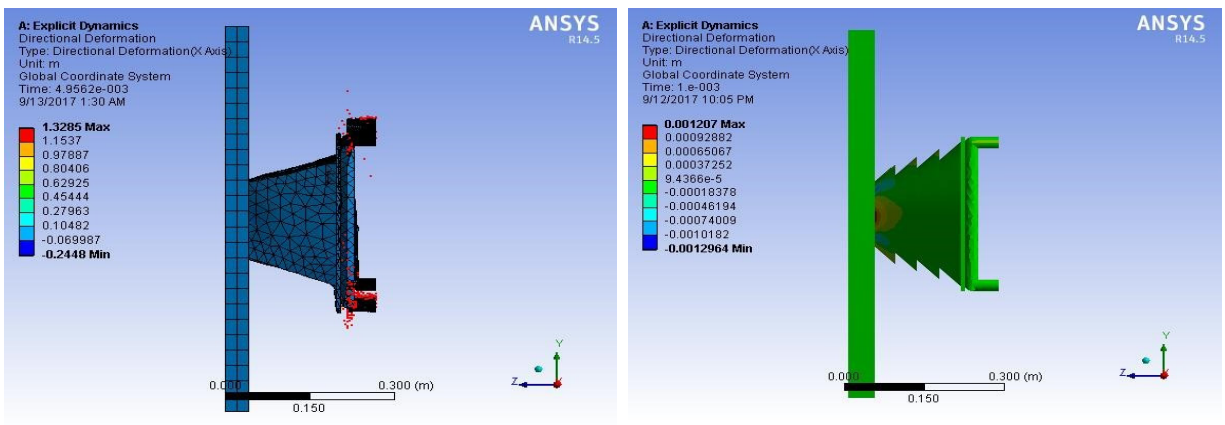


FIGURE 2. Flowchart Simulation Procedure

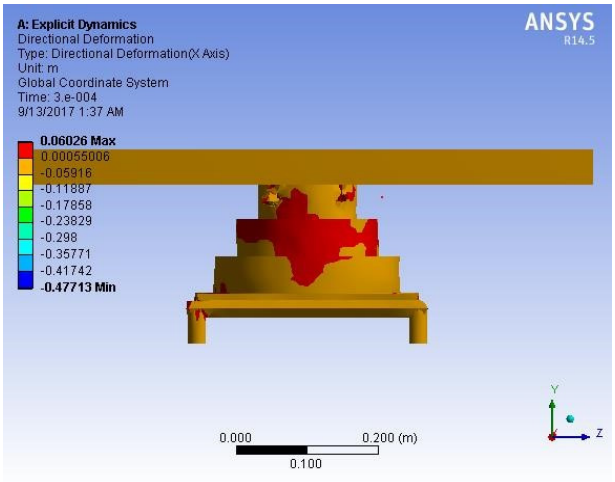
In all simulations performed in all models have the same behavior and circumstances. All models are made up of the same material. Thus, all models have the same properties. In simulating the state of the environment in a steady state in the absence of energy released to the system, the energy all moves only to the elements present in the model. The whole model is also assumed to drive at a constant speed of 7 m/s that starts when it right will hit the rigid wall, so the simulation is run by merely moving the model for 0.01 Second. If the simulation runs until the deadline, then each model will move as far as 700 mm hit the rigid wall. Then the variables included only consist of the speed and properties of each model. With these two variables, the simulation will look for the significant deformation that occurs (deformation to time and deformation to distance). Further, the results to be achieved are the absorbed energy and its impact on each model assessed from the deformation that occurs.

DYNAMIC SIMULATION RESULTS OF IMPACT ATTENUATOR

All models inputted to ANSYS software for simulation process. The input process was done, through the conditions and variables that want to be differentiating factors that will be the assessment of the best model simulation results. By including deformation as one of the variables generated from the simulation, the deformation experienced by each impact attenuator shown in Figure 3.



a. b.



c.

FIGURE. 3 Deformation of Impact Attenuator: (a) Model 1, (b) Model 2, and (c) Model 3

Figure 3 presents the simulation result of deformation pattern of all models. In the first model the Impact attenuator is not significantly deformed, but when viewed on the frame there condition shown with a red dot that signifies the frame of model 1 gives the transfer of impact to the frame, so the frame is damaged. In model 2, the impact attenuator and frame have the same color and are still within the safe boundary of deformation, while in the farthest impact part of the frame is damaged, that is a considerable deformation, it shows that model 2 has a proper impact distribution, and does not harm the frame. In Model 3, both impact attenuators and frames have substantial deformation, and in impact attenuators, there is damage that makes impact attenuator destroyed, so the third model has a proper impact distribution, but the ability to absorb the impact is terrible so that Impact attenuator is damaged and the frame is deformed large enough.

Simulation of impact attenuator presents the transformation of deformation per time. Result deformation per time shown by the graph in Figure 4.

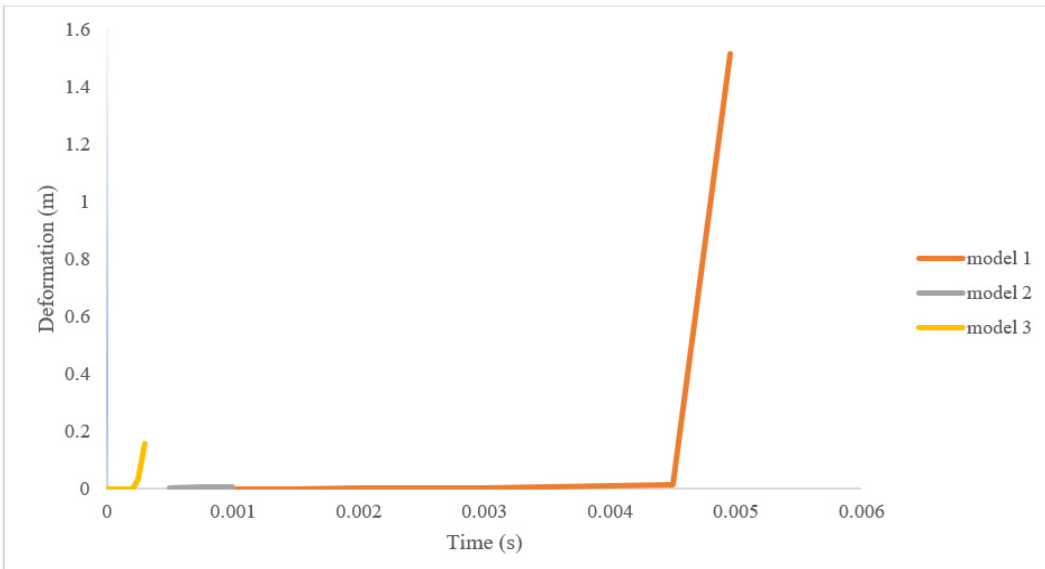


FIGURE 4. Graphic Deformation per Time

Based on the simulation result of deformation per time deliver several things. All models of the impact attenuator have different deformation changes per time. The first model has minimal deformation until the simulation ends, but when compared with the picture, that when the end of the frame simulation is damage in the form of the appearance of fractions. Based on the image results presented the energy absorption of the model 1 defective, because of the energy transfer to the frame. Model 2 has slightly, but regular deformation, from the simulation graph, ended near time 0.001 second which at the same time the model has reached the required target without experiencing much deformation, when compared to the image then model 2 is safe to use. Model 3 undergoes regular deformation at the beginning, but when close to 0.0002 seconds there is a significant deformation close to 0.0003 Second the simulation stops because the impact attenuator is damage in the form of rupture, and then this model cannot continue the simulation.

The next result of the simulation in the form of energy changes over time. The energy absorption per time is presented in Figure 5. Model 1 can reach 7350 J when it is moving at 0.00413 seconds. Model 1 can still absorb the energy without deformation exceeding 25.4 mm. Model 2 can reach 7350 J when it moves to 0.000864 seconds. Even though, model 2 can absorb impact energy up to 250,000 kJ, without deformation more than 25.4 mm as it is close to 0.001 Second. Moreover, model 3 can reach 7350 J when

it moves for 0.000241 seconds. When model 3 reaches 7350 J, it has been deformed beyond 25.4 mm. Therefore, model 3 does not reach the target according to the Formula Student SAE rules.

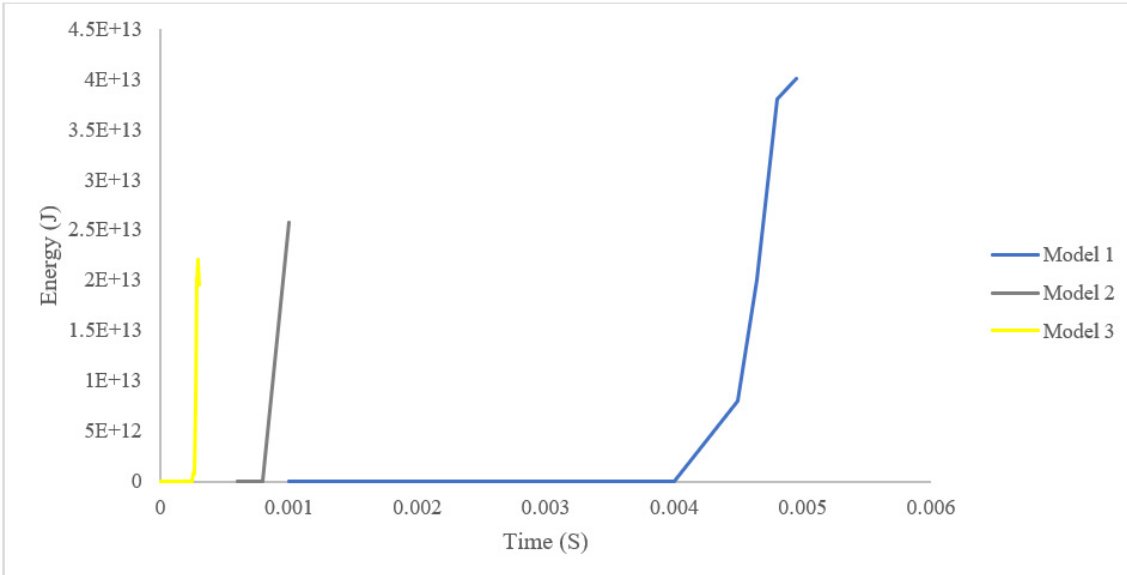


Figure 5. Graph Energy Absorption per Time

CONCLUSION

Several models of impact attenuator have been simulated using ANSYS. According to FSAE rules, the Impact Attenuator model should be able to absorb 7350 J without causing excessive deformation on the frame more than 25.4 mm. The impact attenuator model 1 and 2 could follow the criteria. Based on the energy absorption ability and the Deformation comparison of energy absorption, model 2 is a model of having the best capability. Model 2 can absorb 7350 J of energy with the smallest deformation. Therefore, based on the simulation result, the preferable Impact attenuator model for the current race car is model 2. Last, but not least, model 3 cannot be adopted for the race car due to its performance.

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